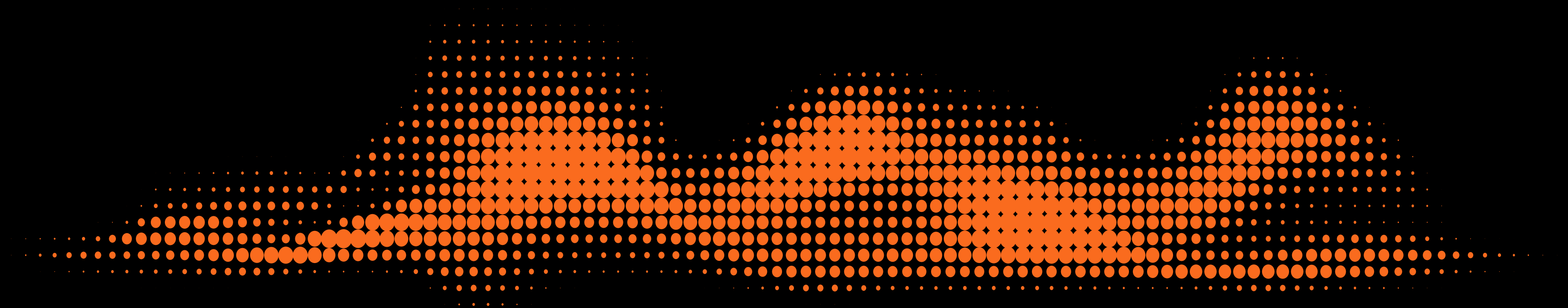




# ActiveMesh, Failover & Control Plane

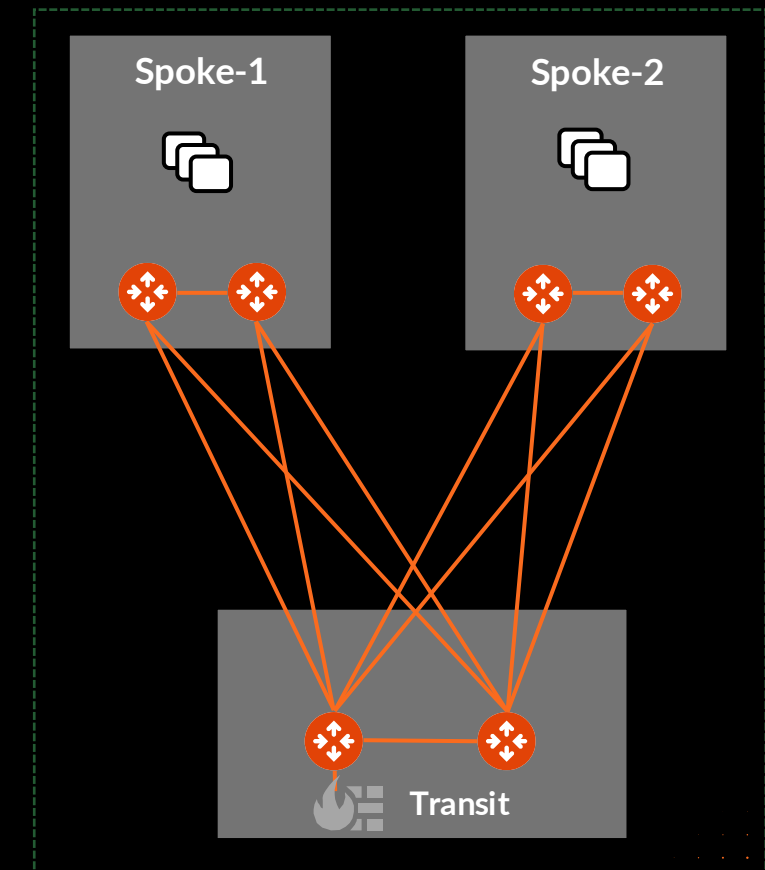


# Overview



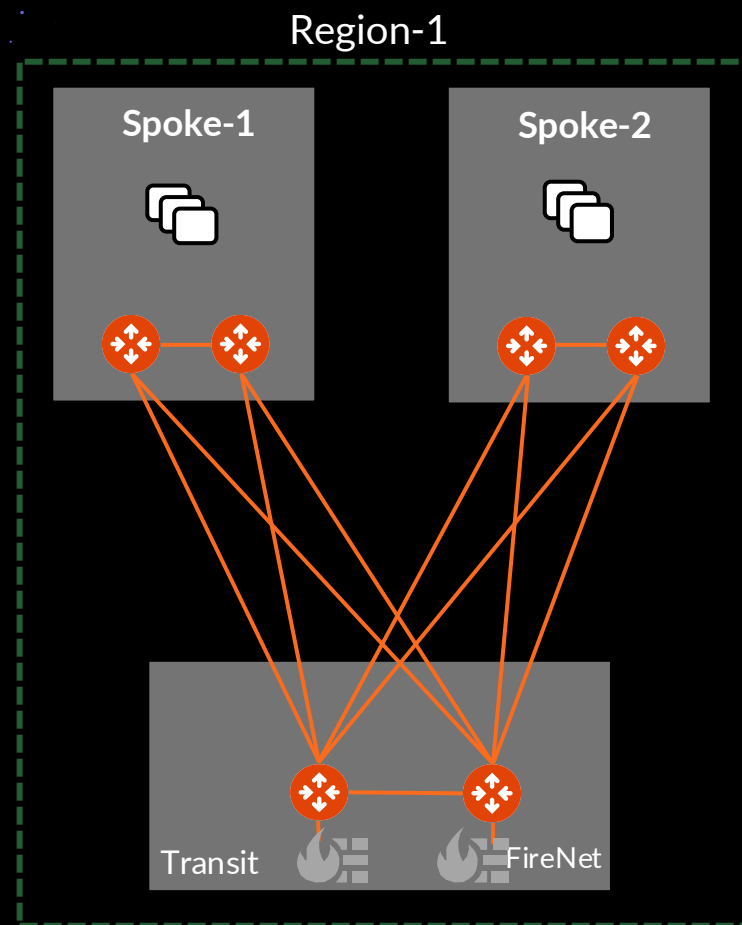
# What is ActiveMesh?

- ❑ Provides network **resiliency**, improved convergence time and high performance
- ❑ Two Aviatrix gateways in a VPC/VNet/VCN form a cluster
- ❑ Both gateways forward traffic simultaneously via ECMP
- ❑ Each gateway in a Spoke VPC/VNet/VCN builds IPsec tunnels to **both** Transit gateways
- ❑ Number of Transit and Spoke gateways as well as their **instance sizes** are independent of each other:
  - Maximum **15x** Transit Gateways can be deployed per Transit VPC/VNet/VCN
  - Maximum **15x** Spoke Gateways can be deployed per Spoke VPC/VNet/VCN

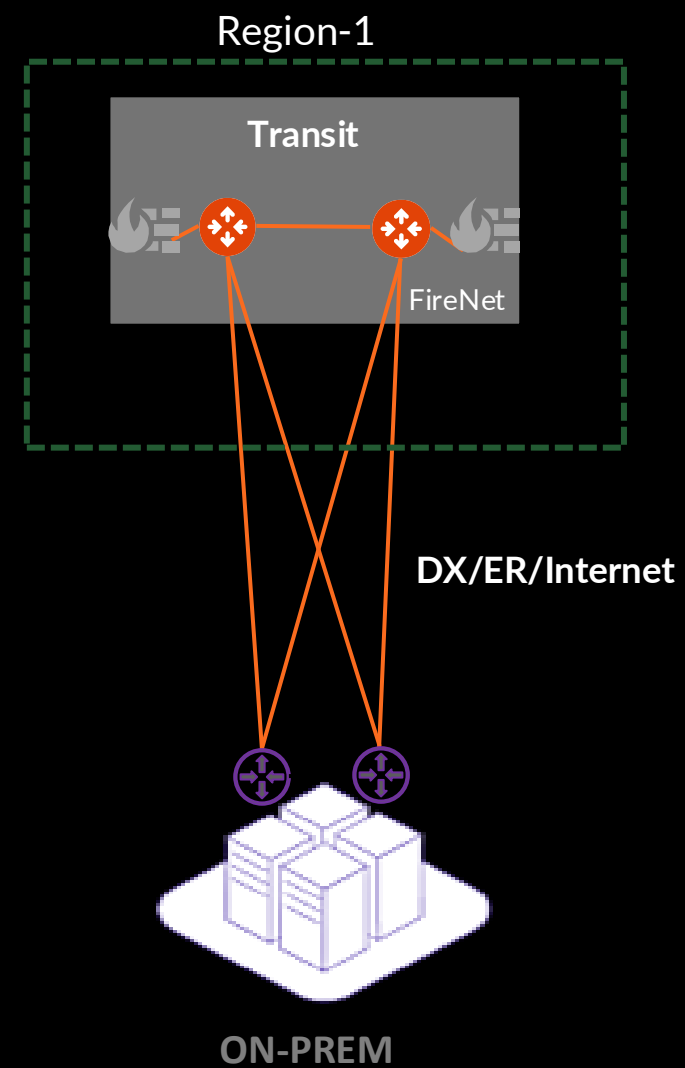


# Use Cases

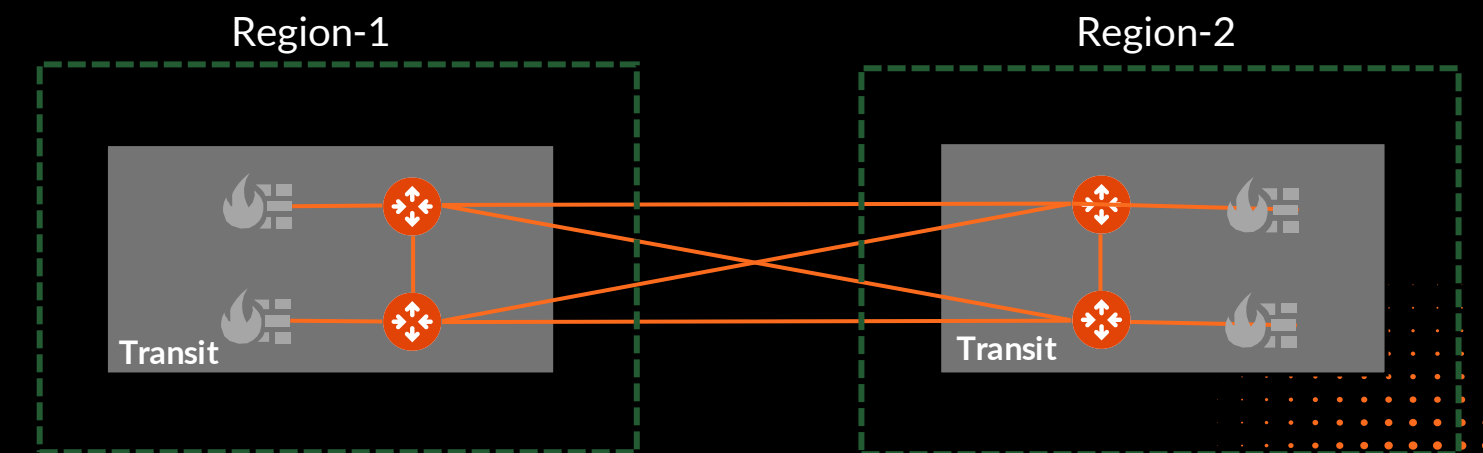
## Intra-Region Spoke-Spoke



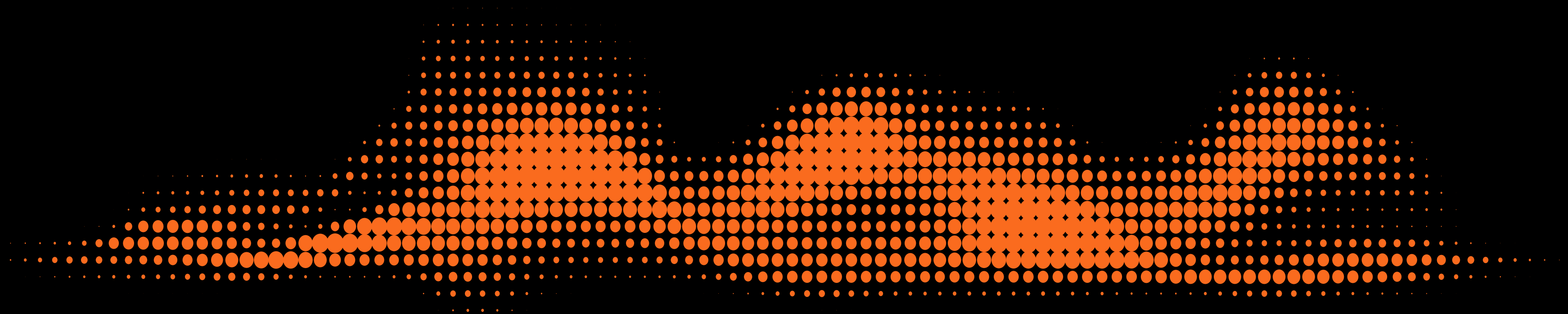
## Cloud to On-Prem



## Inter-Region / Multi-Cloud



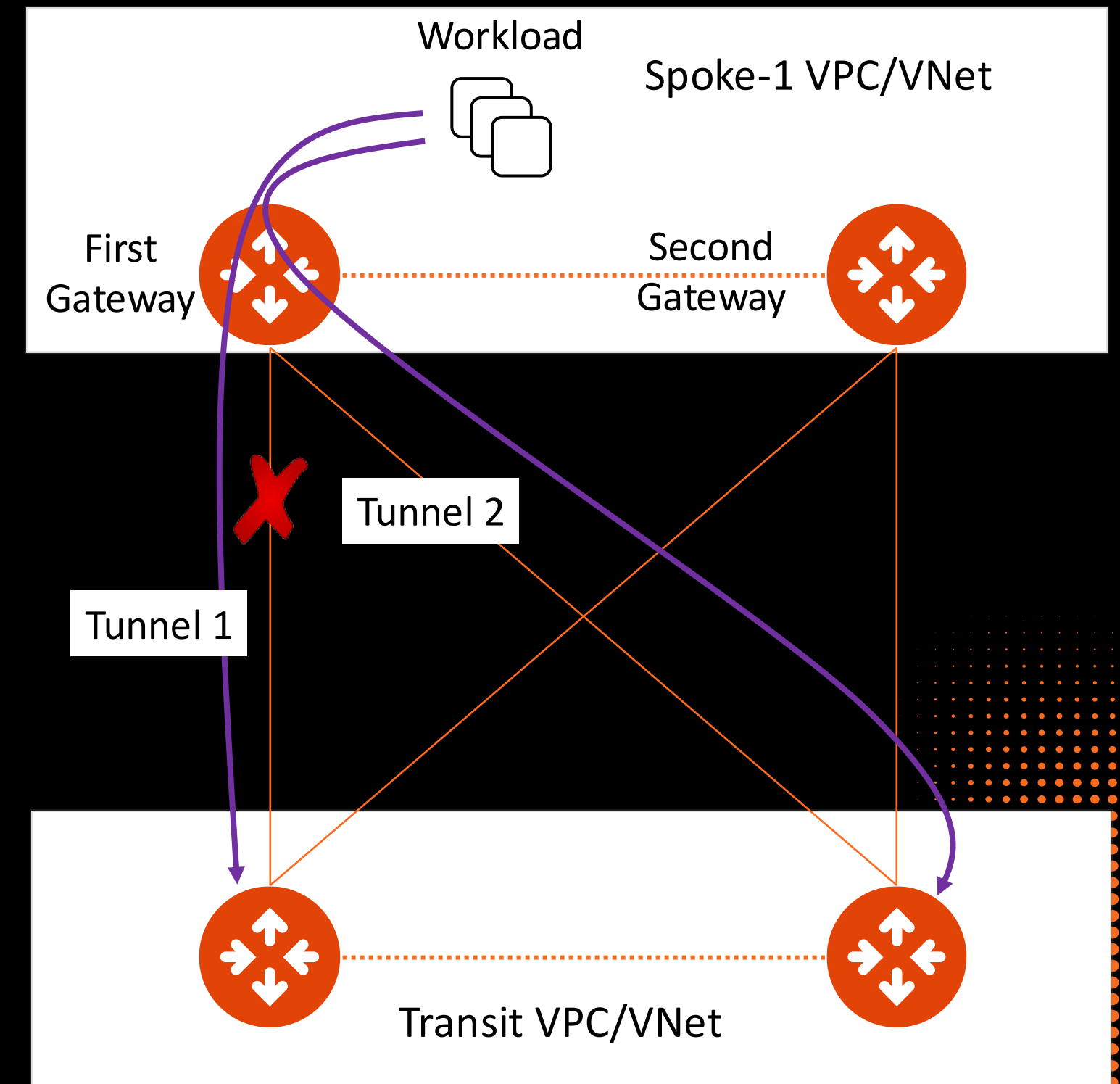
# Failover





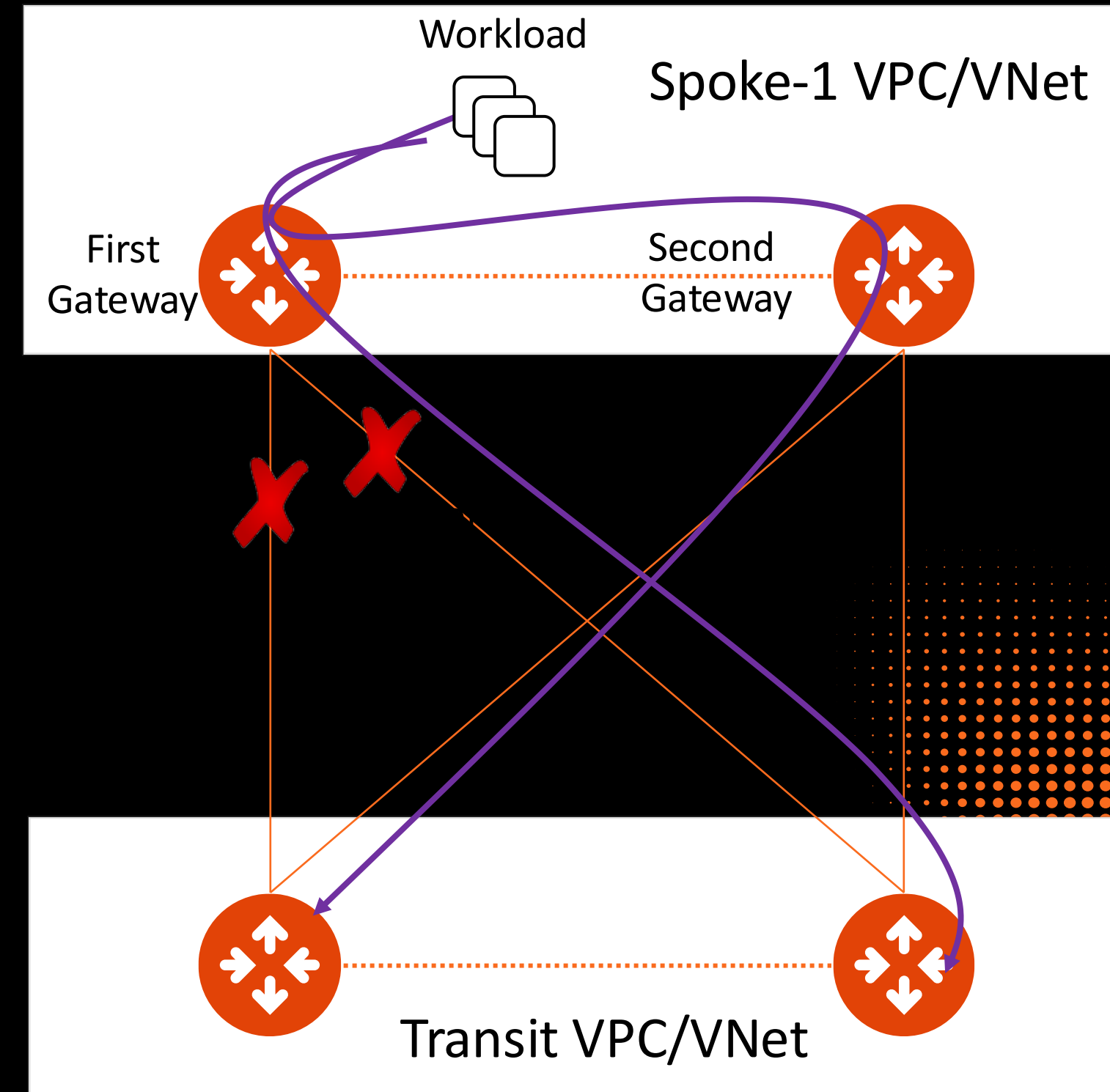
# Failover Scenario 1

- ❑ Workload in Spoke-1 VPC/VNet: Primary gateway → Tunnel 1 → Transit → Spoke-2 VPC/VNet (not shown).
- ❑ If Tunnel 1 at the first Spoke Gateway fails, traffic uses Tunnel 2 to the second Transit Gateway.
- ❑ Tunnel 2 was already active, forwarding half the traffic (**metric 100**).
- ❑ No route re-convergence in the VPC/VNet route table.
- ❑ Gateway self-manages the failover
- ❑ The controller is informed of the tunnel-down event but does not perform the change.



# Failover Scenario 2

- ❑ Workload in Spoke-1 VPC/VNet traverses the Primary gateway via Tunnel 2 to the Transit toward Spoke-2 VPC/VNet (not shown).
- ❑ If both Spoke-to-Transit tunnels fail at the Primary Spoke gateway, traffic is forwarded through the interconnected link to the Secondary Spoke gateway (**metric 200**).
- ❑ The Secondary Spoke gateway sends traffic to the Transit Gateways via ECMP (metric 100 on both downstream links).
- ❑ No route re-convergence occurs in the VPC/VNet route table.
- ❑ The gateway autonomously handles the change
- ❑ The controller is aware of the tunnel-down event but does not make the change.





# Failover Scenario 3

AZ-A

route table RT-A

Initially: 10.0.0.0/8 → Spoke-AZ-A-GW NIC

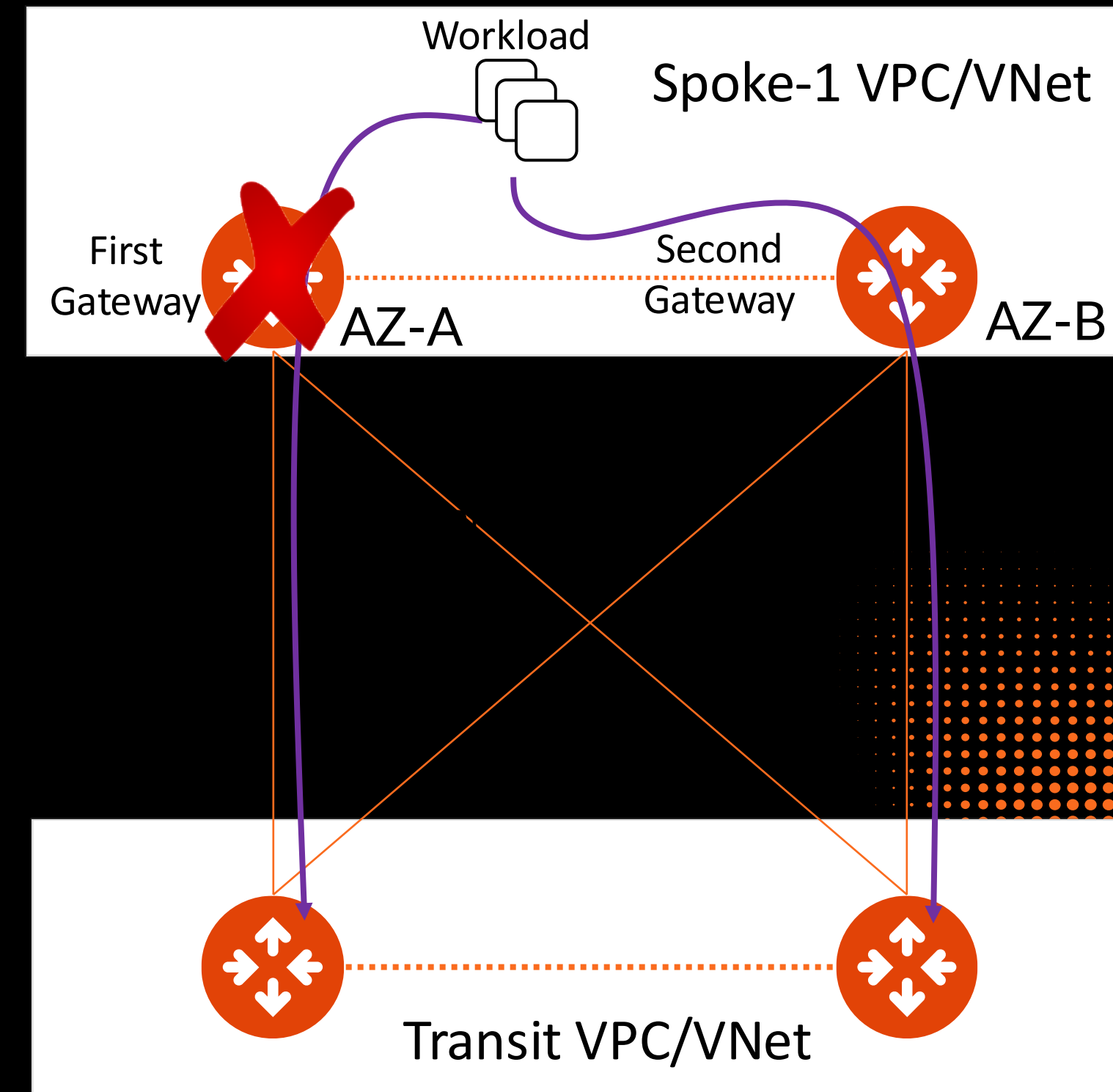
After failover: 10.0.0.0/8 → Spoke-AZ-B-GW NIC

AZ-B

route table RT-B

10.0.0.0/8 → Spoke-AZ-B-GW NIC

- ❑ The Spoke-1 VPC/VNet workload requires reachability to Spoke-2 VPC/VNet (not shown); however, the gateway is unavailable.
- ❑ Upon failure of the first gateway, the controller detects the event via *periodic keepalive messages* exchanged with the gateways.
- ❑ The controller then **reprograms** the routing table in AZ-A, updating the next hop for the three RFC1918 routes to the ENI of the Secondary Spoke Gateway in AZ-B.





# Active-Standby Mode

- ❑ Use case: Deployment scenario where an on-prem device such as a firewall does not support asymmetric routing on two tunnels
- ❑ Upon failure, the Second gateway takes over from the First gateway
- ❑ The first does not become active unless there is a manual switchover or the Second fails
- ❑ UI provides an option for customers to choose **Preemptive** or **Non-Preemptive** behavior.

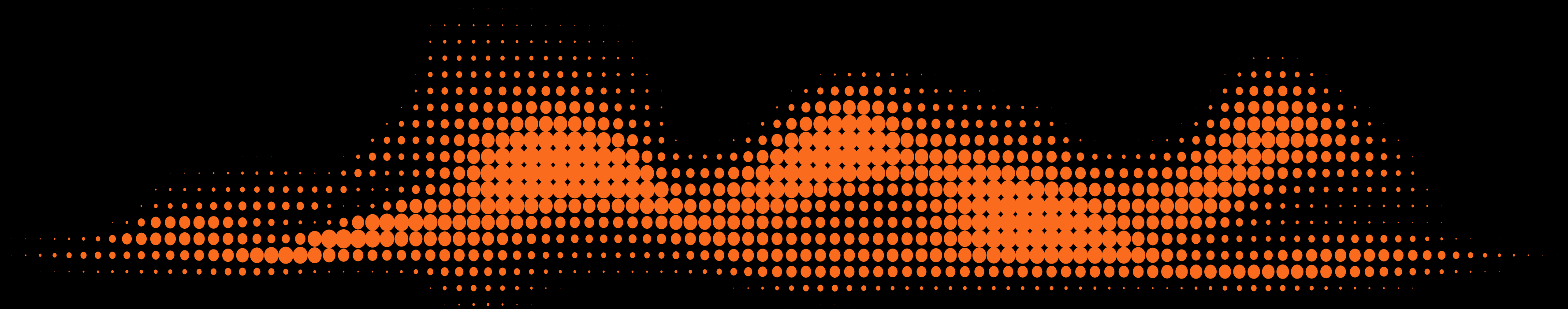
The screenshot shows the AWS Transit Gateway console for the instance 'AVX-AWS-TRANSIT-GW'. The 'Settings' tab is selected, and the 'General' section is expanded. The 'Active-Standby' toggle is highlighted with a red box and is currently set to 'Off'. Other settings include 'Use VPC/VNet DNS Server' (Off), 'Connected Transit' (On), 'Advertise Transit VPC/VNet CIDR' (Off), 'Multi-Tier Transit' (Off), 'Jumbo Frame' (On), 'GRO/GSO' (On), and 'Gateway Single AZ HA' (On). The 'Change Interface(s) RX Queue Size' section is also visible.

The modal dialog for configuring Active-Standby mode is shown. The 'Active-Standby' toggle is turned 'On'. Under the 'Failover Mode' section, the 'Non-Preemptive' radio button is selected, indicated by a red arrow and the word 'Default'. The 'Preemptive' radio button is unselected. The dialog includes 'Cancel' and 'Save' buttons.

# Control Plane

## SD Routing

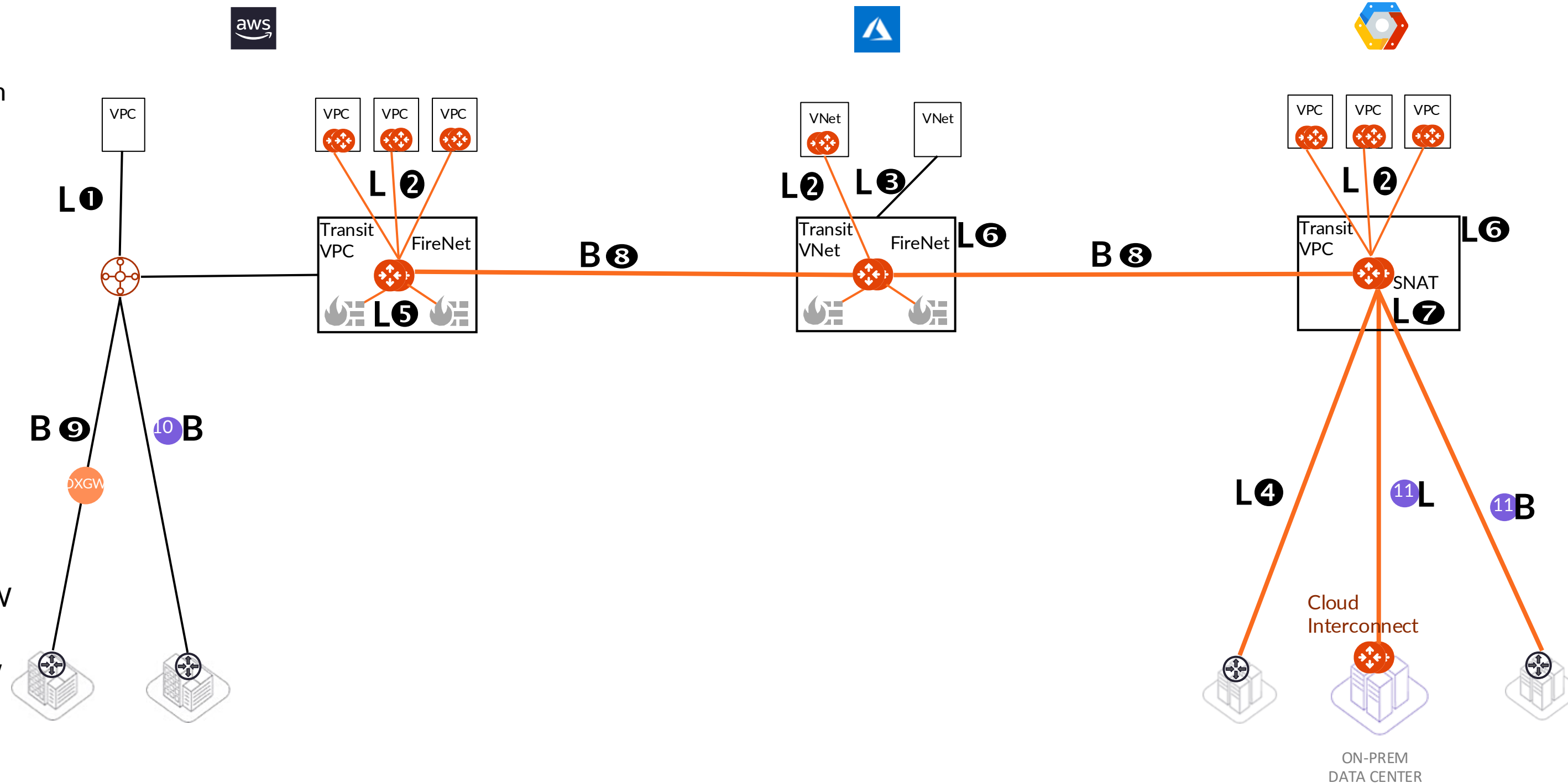
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# Route Classification

1. AWS TGW Attachment [L]
2. Aviatrix Spokes (VPC/VNet) [L]
3. Azure Native Spokes [L]
4. Aviatrix Transit GW – on-prem (*static*) [L]
5. Firewall Egress 0/0 [L]
6. Transit VPC/VNet associated prefixes [L]
7. Transit GW SNAT IP [L]
8. Remote Transit GW (Transit Peering) [B]
9. TGW DXGW [B]
10. TGW VPN [B]
11. Site2Cloud BGP on Transit GW (including Edge routes) and Site2Cloud BGP on Spoke GW [B]

L = Routes considered local by Controller  
B = BGP learned routes



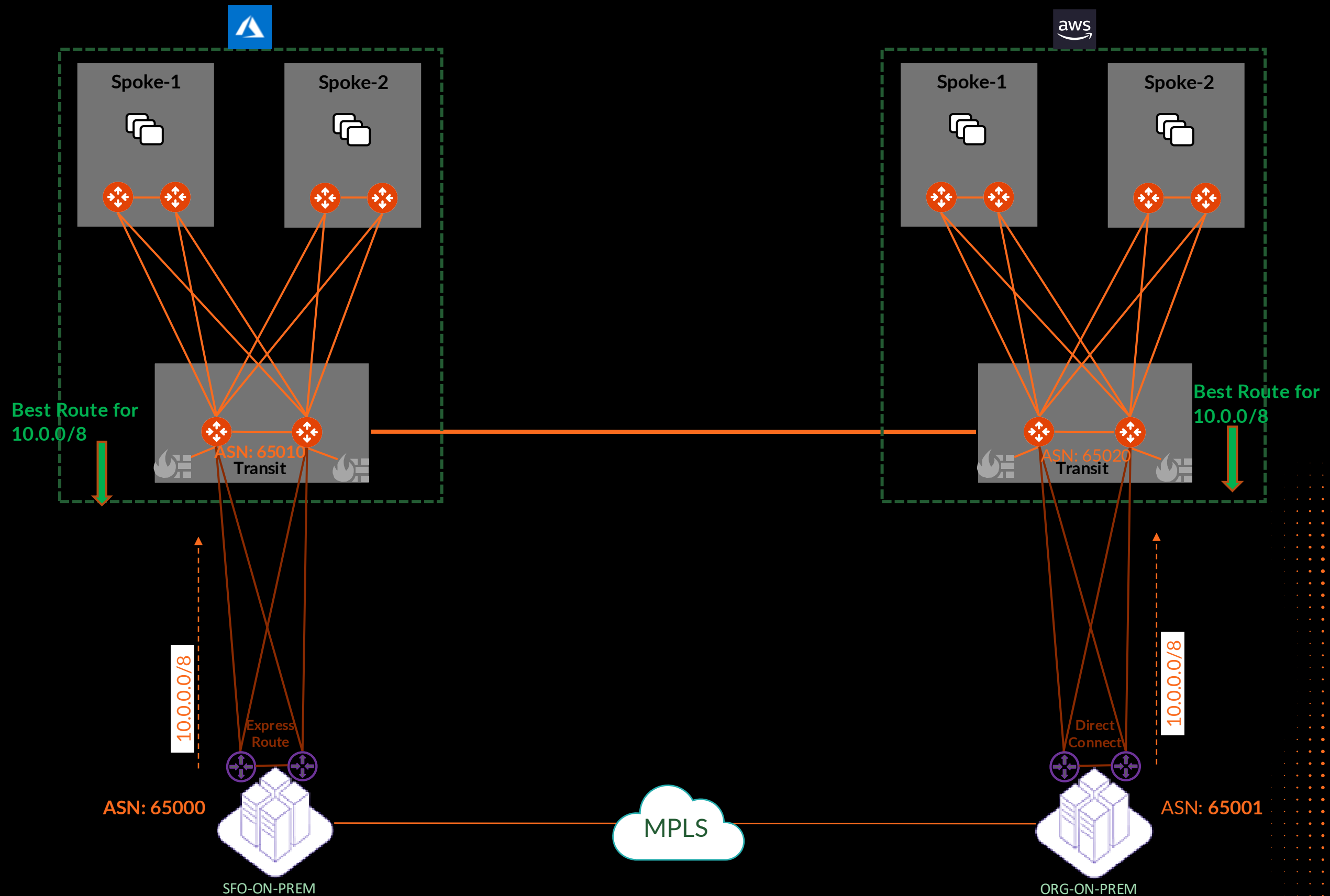
1. Longest prefix match
2. If equal length, then local route is chosen
3. If routes are of the same type, then shortest AS-path length is chosen
4. If AS-path length is the same, then lowest metric is chosen
5. If metric is the same, then
  - If ECMP is enabled, then traffic is distributed to available routes
  - If ECMP disabled, then the route first programmed in the table is chosen
  - If programmed at the same time, then lower integer IP next hop is chosen



# Example of using Transit as an alternate path 1/3

## At steady state

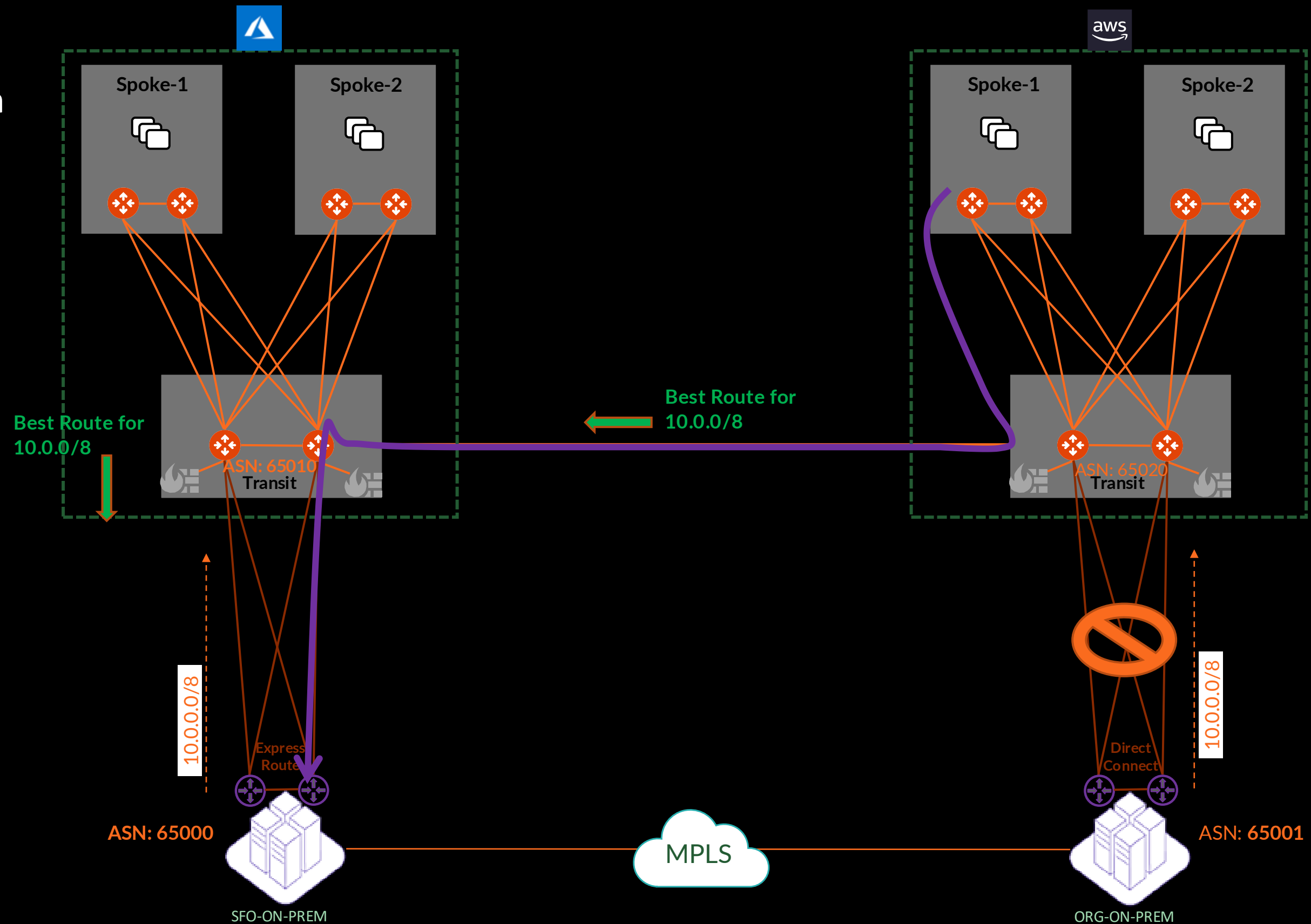
- Each transit is learning 10/8 locally from on-prem
- For each transit, Controller DB will have 10/8 via local and peer transit
- Route via peer will have as-path-len 2
- Each transit and its spokes will get to on-prem via local private path (DX/ER) as best path





# Example of using Transit as an alternate path 2/3

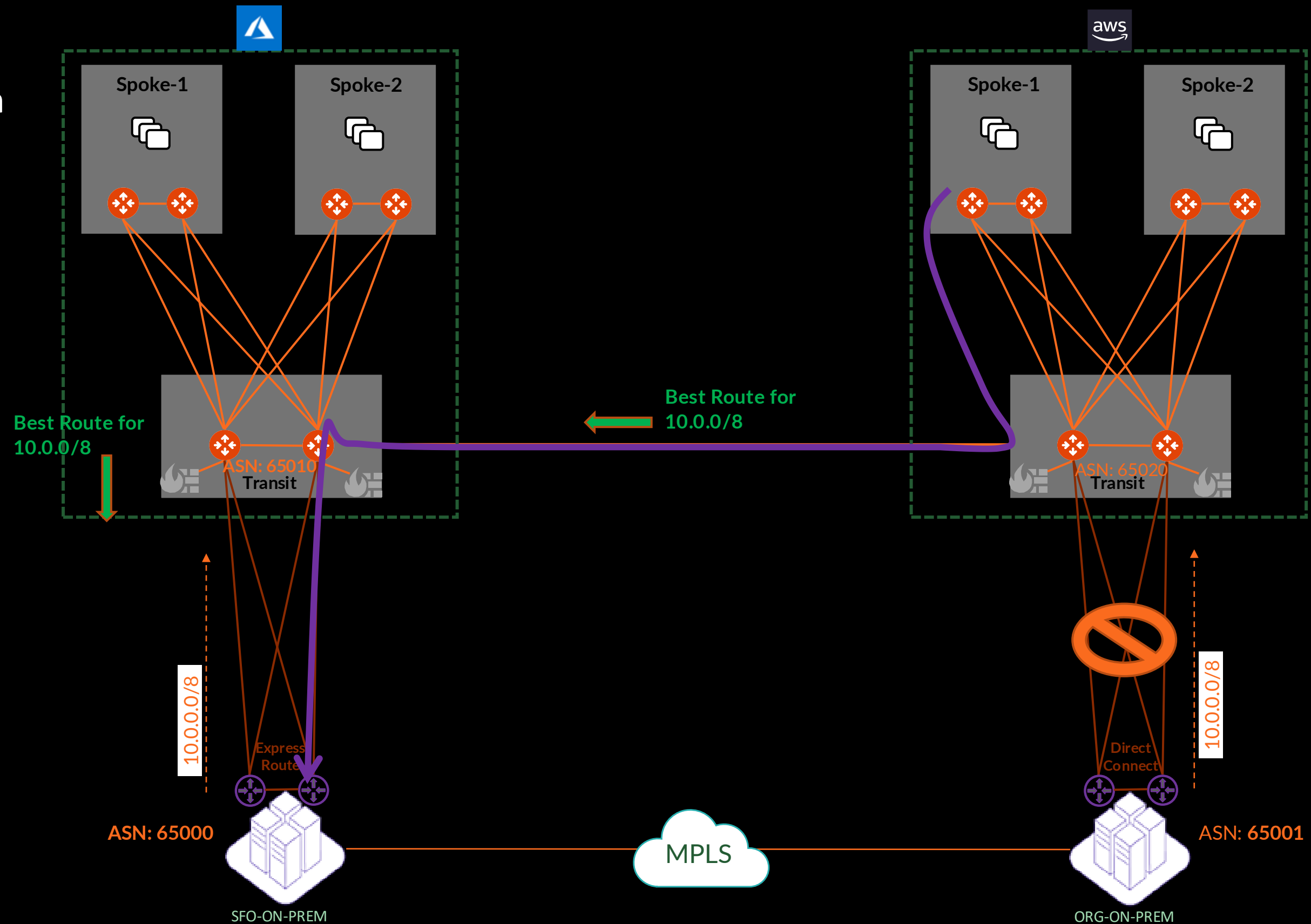
- ❑ When on-prem connection goes down.
  - For e.g., DX is down.
  - Only route to 10/8 now is via Azure Transit.





# Example of using Transit as an alternate path 2/3

- ❑ When on-prem connection goes down.
- For e.g., DX is down.
- Only route to 10/8 now is via Azure Transit.

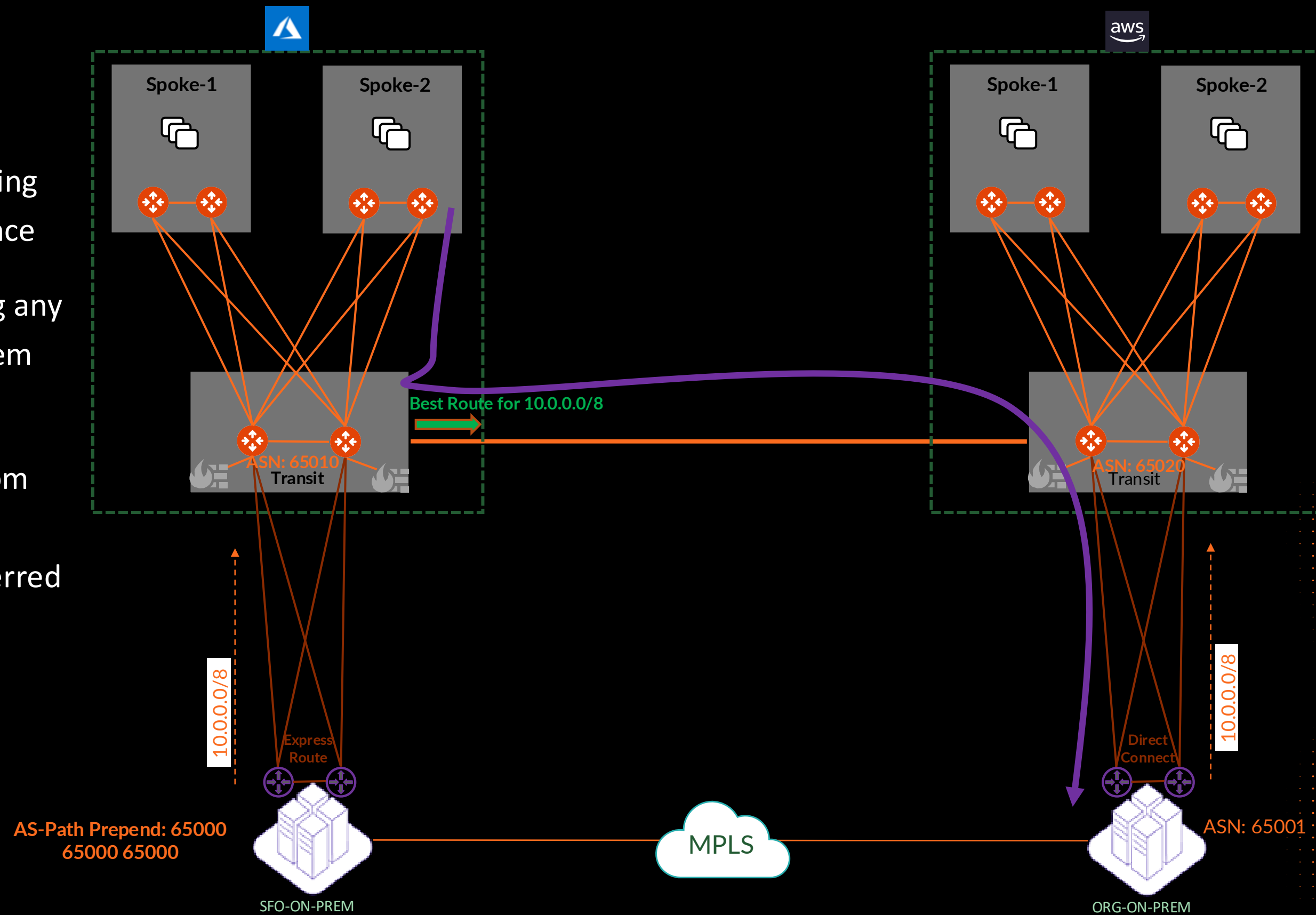




# Example of using Transit as an alternate path 3/3

## Use AS-PATH Prepend

- E.g, SFO on-prem ER is going under planned maintenance
- You want to avoid sending any traffic through SFO on-prem ER
- You can send AS-paths from SFO on-prem so that AWS Transit becomes the preferred path



# Next: Lab 4 - HPE

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